
team1
BUILDERS
BRUNCH



avalanche

Sealed-Bid Auctions on L1

Part 2 - Advance Level

Who is in the Room

Hi 👋 I'm

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Collaborator @ team1 avalanche



@0xdev8

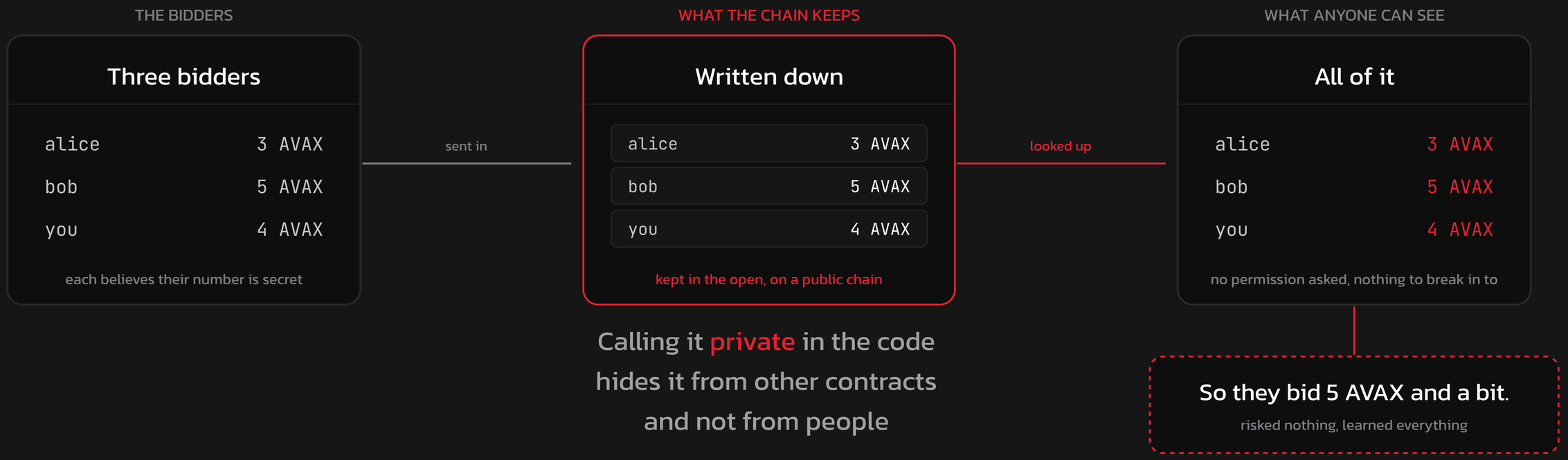
ahmedali

ahmedali.dev

ate

Recap: The Naive Auction.

Anyone could read every bid



What if we ran it on our own chain?

Same Code. Different Room.

THE CONTRACTS

NaiveAuction and SealedBidAuction

Fuji C-Chain	
send a transaction	anyone
deploy a contract	anyone
gas token	AVAX
base fee	market price
read every slot	anyone

the whole world is in the room

Our L1 · chain 1717	
send a transaction	invited only
deploy a contract	the admin
gas token	BRUNCH
base fee	1 wei / gas
read every slot	everyone inside

only the people we let in

The room got smaller. Everyone in it still reads every slot.

The Door.

OUR L1 · CHAIN 1717

Only invited accounts can send a transaction.

OUTSIDE

Carol

balance	1000 BRUNCH
transact	none
deploy	none

she can pay, nobody invited her

cannot issue transaction
from non-allow listed address

bid 5 BRUNCH

THE DOOR

Transaction allowlist

precompile at
0x0200…0002

set in genesis,
changed by the admin

on the list

INSIDE

Invited

admin	admin
alice	enabled
bob	enabled

1000 BRUNCH each

just allow carol

and she's in, one transaction by the admin

She can pay. She can't get in.

The Room.

INSIDE THE ROOM

Everyone here was invited. Alice and Bob both bid.

ALICE

Alice bids

allowlist	enabled
balance	1000 BRUNCH
bid	3 BRUNCH

she believes her number is secret

sent in

WHAT THE CHAIN KEEPS

Written down

contract	NaiveAuction
slot	0x14e0…d101
value	3 BRUNCH

private and in storage

read

WHAT BOB CAN SEE

Bob reads it

allowlist	enabled
ran	just peek
alice bid	3 BRUNCH

no transaction, nothing to stop

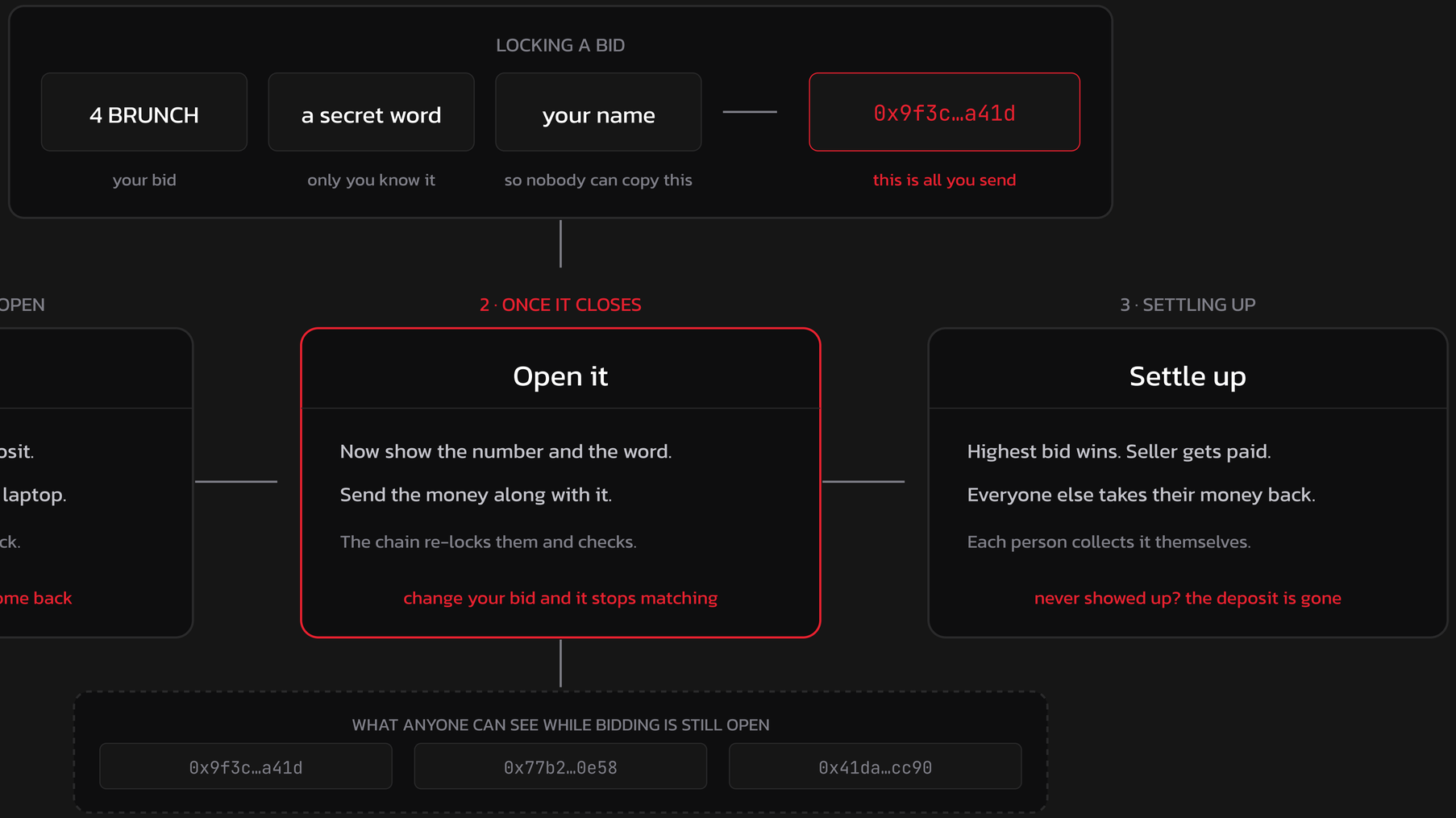
So Bob bids 3 BRUNCH and a bit.

same attack as part 1, from inside

The allowlist guards transactions. Reading storage isn't one.

Commit–Reveal. Still Needed.

Lock your bid first. Open it once bidding has closed.



What It Bought. And what it didn't.

Our own chain bought

- ✓ who sends transactions
- ✓ who deploys contracts
- ✓ who validates
- ✓ the fees

It did not buy

- ✕ secrecy from anyone inside
- ✕ a reason to drop commit-reveal

in a tender, the people inside are your competitors

ALLOWLIST
who is in the room
today

COMMIT-REVEAL
hidden until the reveal
part 1 · still needed

ENCRYPTION · eERC
hidden for good
next session

Who is in the room. What they can see. For how long.

Questions?

Thank you!

@AvaxTeam1

@0xdev8